

SENTINEL AI — DEEPPFAKE INVESTIGATION REPORT

INVESTIGATION ID: CASE-TEST-M9 — DIGITAL EVIDENCE EXAMINATION

1. INVESTIGATION METADATA

Investigation ID:	CASE-TEST-M9	Case Status:	CASE_UNDER_INVESTIGATION
Case Title:	Milestone 9 PDF Report Engine Test	Evidence Items:	1
Investigator / Owner:	Test User	Assigned Expert:	Test User
Incident Date:	N/A	Analysis Date:	2026-09-24 16:53 UTC
Created Date:	2026-09-24 22:00 UTC	Model Version:	Sentinel AI V1.7-A Dual-Head

2. EXECUTIVE SUMMARY

Total Evidence Analyzed	Tampered Classifications	Authentic Classifications	Localization Maps
1	1	0	1 / 1

3. DETAILED EVIDENCE EXAMINATION

EVIDENCE ITEM E-001: evidence_m9_test.jpg

Evidence ID:	15	MIME Type:	image/jpeg
File Size:	41.20 KB	Upload Timestamp:	2026-09-24 16:53 UTC
Analysis Timestamp:	2026-09-24 16:53 UTC	SHA-256 Hash:	57244f424f284ffbb6aae62425f06ea5000000 00000000000000000000000000000000



Figure 1: Original Evidence Image



Figure 2: Sentinel AI Heatmap Overlay

Metric	Sentinel AI V1.7-A Value	Threshold Criteria
Classification Result	TAMPERED / MANIPULATED MEDIA	Threshold: 0.40
Tampered Probability	75.0%	Range: [0.0% - 100.0%]
Authentic Probability	25.0%	Range: [0.0% - 100.0%]
Spatial Localization	Heatmap Artifact Generated	Threshold: 0.35

4. MODEL METHODOLOGY & TECHNICAL DETAILS

Architecture Overview: Sentinel AI V1.7-A is a frozen dual-head deep learning architecture optimized for image deepfake detection and pixel-level spatial localization.

Preprocessing Stream: Input images are normalized and passed through a dual-stream RGB backbone (EfficientNet-B0) paired with a high-pass Laplacian residual filter bank capturing high-frequency forensic noise signatures.

Dual Heads:

- *Classification Head:* Computes softmax probability across authentic vs. tampered classes using a calibrated decision threshold of **0.40**.
- *Localization Head:* Generates pixel-level feature heatmaps highlighting manipulated regions using a spatial activation threshold of **0.35**.

5. FORENSIC INTERPRETATION & DISCLAIMER

OFFICIAL FORENSIC DISCLAIMER

This report presents automated deepfake classification and spatial localization results produced by Sentinel AI V1.7-A. These findings provide probabilistic indicators based on forensic noise, boundary inconsistencies, and facial feature distributions. This document is intended to assist law enforcement, legal experts, and digital forensic investigators as corroborative evidence. Final evidentiary conclusions should incorporate qualified human expert examination.